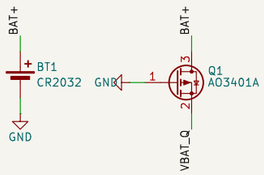
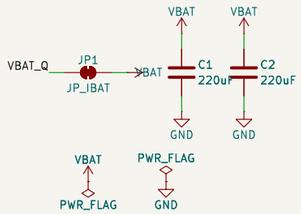
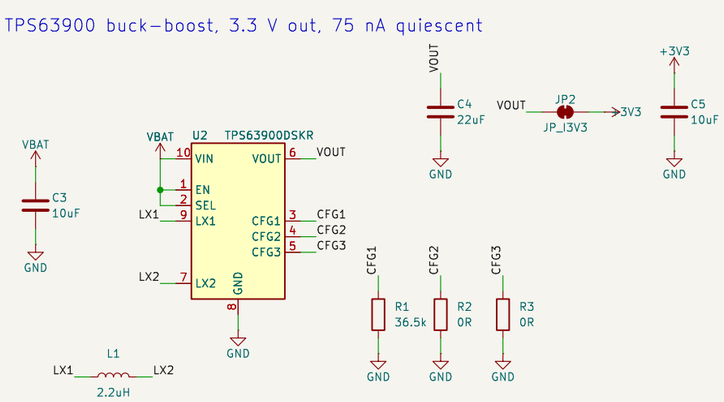
1 Battery CR2032, reverse polarity protection, bulk capacitors

Q1: drain to the cell, source to the load, gate to GND (an upside-down cell turns it off)





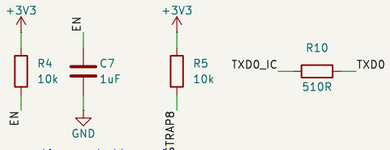
2 TPS63900 buck-boost, 3.3 V out, 75 nA quiescent



CFG1 36.5k + CFG2 0R: VO(2) = 3.3 V, no input current limit; CFG3 0R: VO(1) = 1.8 V, unused: EN, SEL to VIN

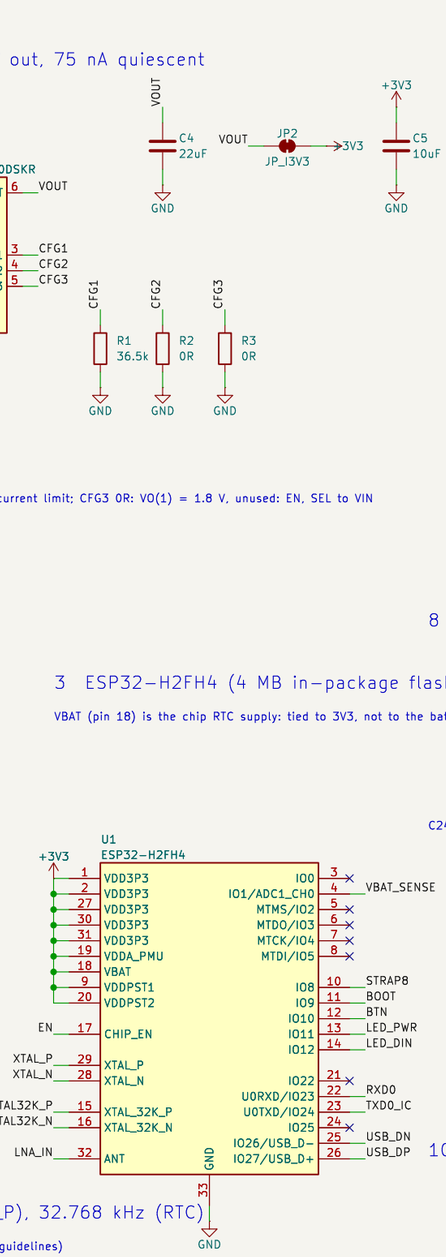
4 Test pads (bottom side, pogo pins)

5 EN delay, strapping, UART0 TX

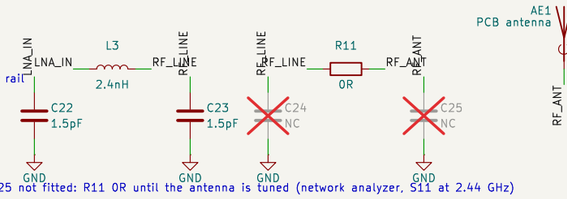


3 ESP32-H2FH4 (4 MB in-package flash)

VBAT (pin 18) is the chip RTC supply: tied to 3V3, not to the battery rail



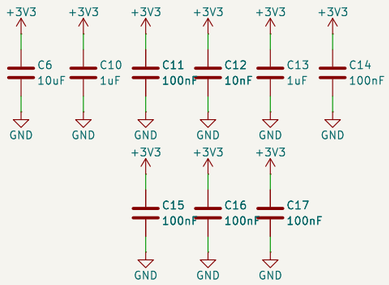
8 RF: CLC at the chip (C22 L3 C23), 50 ohm line, antenna tuning pi (C24 R11 C25)



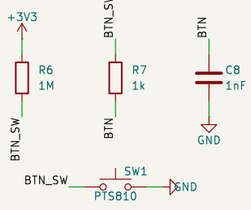
C24, C25 not fitted: R11 0R until the antenna is tuned (network analyzer, S11 at 2.44 GHz)

6 Decoupling at the pins

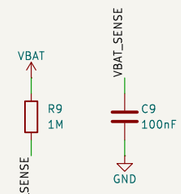
At the pins: C6 C10 C11 1-2, C12 30-31, C13 C14 27, C15 18-19, C16 20, C17 9



9 Button GPIO10 (LP IO)

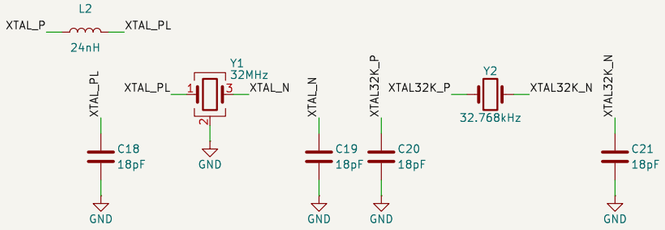


11 VBAT sense GPIO1

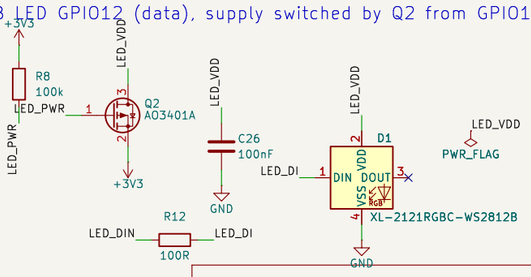


7 Crystals: 32 MHz +-10 ppm (24 nH series at XTAL_P), 32.768 kHz (RTC)

Load capacitors to be trimmed: TX tone frequency offset within +-10 ppm (Espressif guidelines)



10 RGB LED GPIO12 (data), supply switched by Q2 from GPIO11 (low = on)



Generated by scripts/gen_sch.py from scripts/design.py
CR2032 -> Q1 (reverse polarity) -> TPS63900 3.3 V -> ESP32-H2FH4, PCB antenna
Haxeko

Sheet: /		
File: matter-button-cr2032-chip.kicad_sch		
Title: ESP32-H2 Matter over Thread CR2032 button, chip and PCB antenna		
Size: A3	Date: 2026-10-29	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1